Why the LS machine's best window is k=11, and why entropy still does not close the 1D/2D gap

The window is bought with entropy, and copies are the low-entropy trajectories -- but x* goes from 1.92x apart to 496x apart when the entropy prediction is substituted for it. Third refutation.

THE PREDICTION, REGISTERED BEFORE THE RUN. A generation escapes a training set of P strings only if it injects more than ln P nats, so the effective support is exp(H_tot) and x* = exp(H_tot)/|R_L| with no fitted constant.

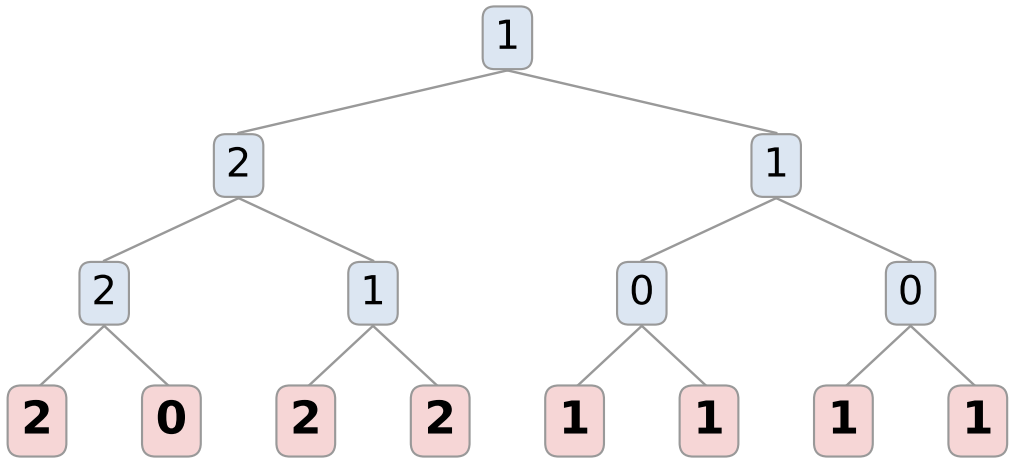
THE RUN. Jobs 51073161 and 51076386. 988 1D rows over 50 bracketed curves, plus all 25 bracketed 2D L=2 cells on the same instrument -- first head-to-head.

CONFIRMED. The mechanism, sample by sample: copies carry 1.75 fewer nats in 1D and 7.42 fewer in 2D, in 84% and 100% of runs. Widening k costs 11.82 -> 8.21 nats over k = 3..17, which is why the optimum is k=11, not U1=15.

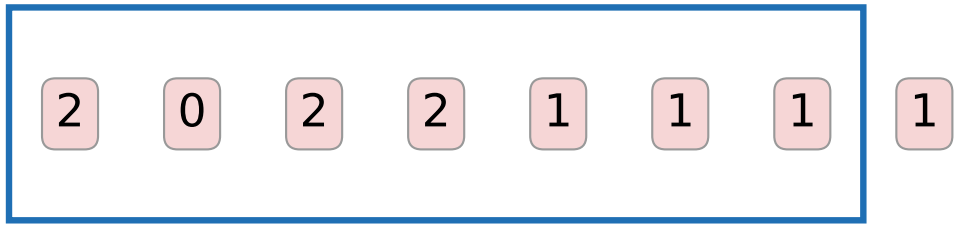
REFUTED. Every quantitative form. novel = exp(-P/exp(H_tot)) misses by 2x in 1D, 600x in 2D; pooled ln P* = 0.100 H + 4.18, slope 0.10 where it needs 1.

1. Tokenization: the string, the tree, the window

v=3, s=2, L=3: root -> 8 leaf tokens



the sample the machine actually sees

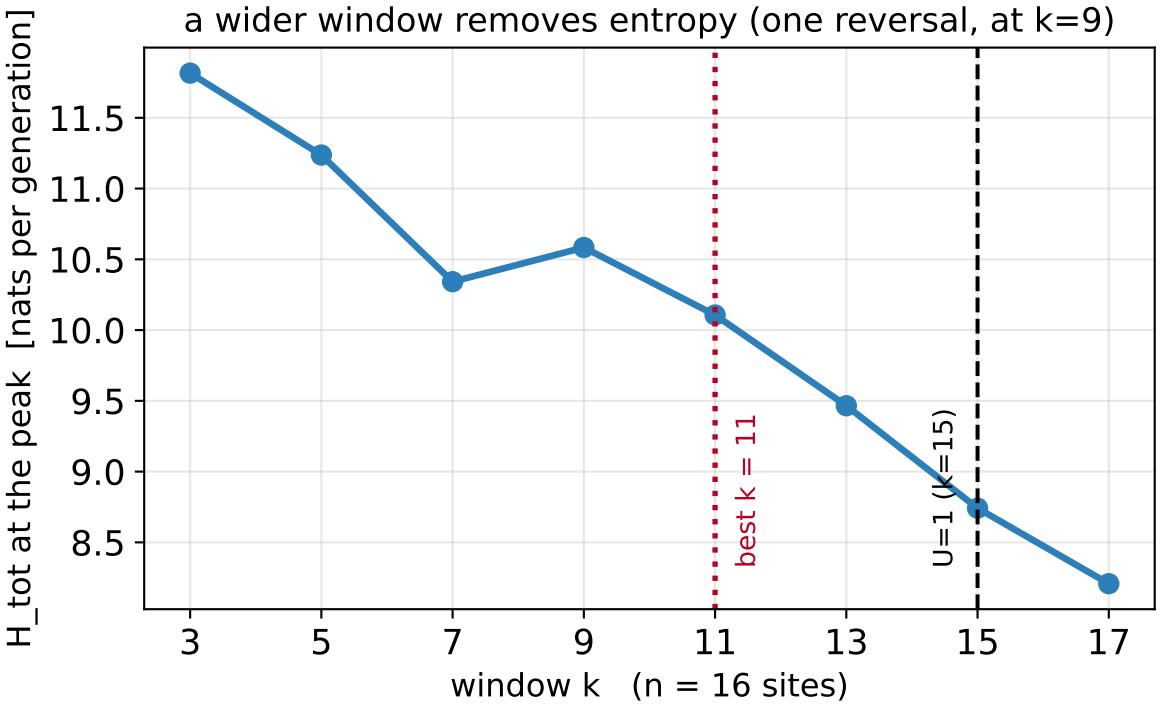


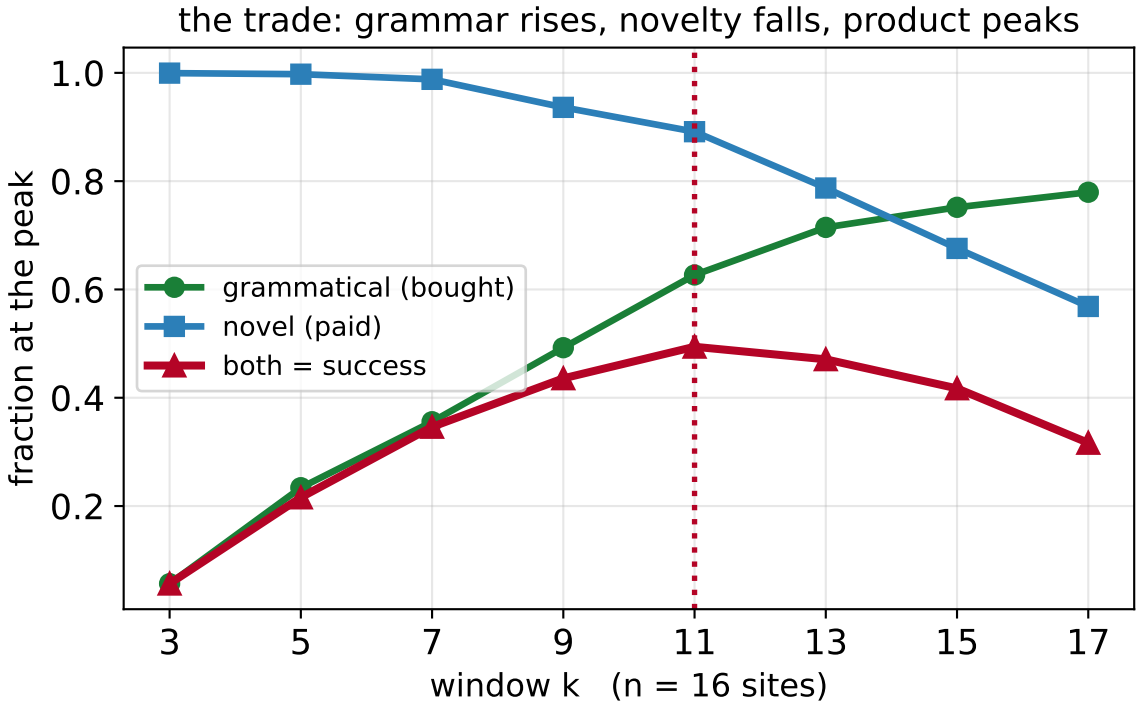
one U=1 window, k = 2s^(L-1) - 1 = 7

covers 7 of 8 sites = 88% of the sample

A token is one leaf symbol from the alphabet 0..v-1. One *sample* is the whole length-d string, d = s^L. The machine is a lookup table over length-k windows: for a masked site it returns the empirical distribution of the centre tokens of every training window that agrees on all observed sites. Off-grid sites get their own marker symbol v and count as observed, so a window at the border can only match training windows framed the same way.
k = 2s^(L-1) - 1 is the smallest window that fully contains the top-level block, i.e. the smallest one that can see the root. In 1D that is d - 1 at every depth, and this deck is about why generating at that window is the wrong choice.

3. Why success peaks at k=11: the window is paid for in entropy



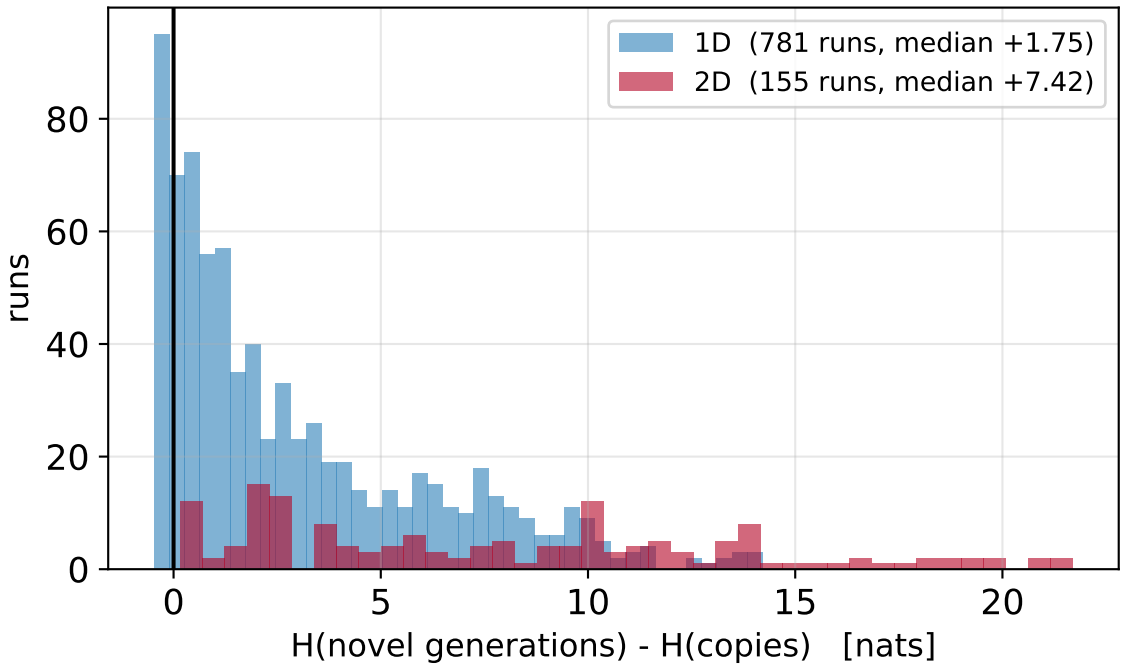


THE LEDGER. Over k = 3..17 at n=16 the trajectory entropy falls 11.82 -> 8.21 nats, a third of what is there, with one reversal (k=7 -> 9) and no feature whatever at U1=15. Grammaticality over the same range rises 0.06 -> 0.78 while novelty at the peak falls 1.00 -> 0.52. The window buys grammar and pays in entropy, and entropy is the currency novelty is denominated in -- so the optimum is wherever the marginal grammar gain stops covering the marginal novelty loss.

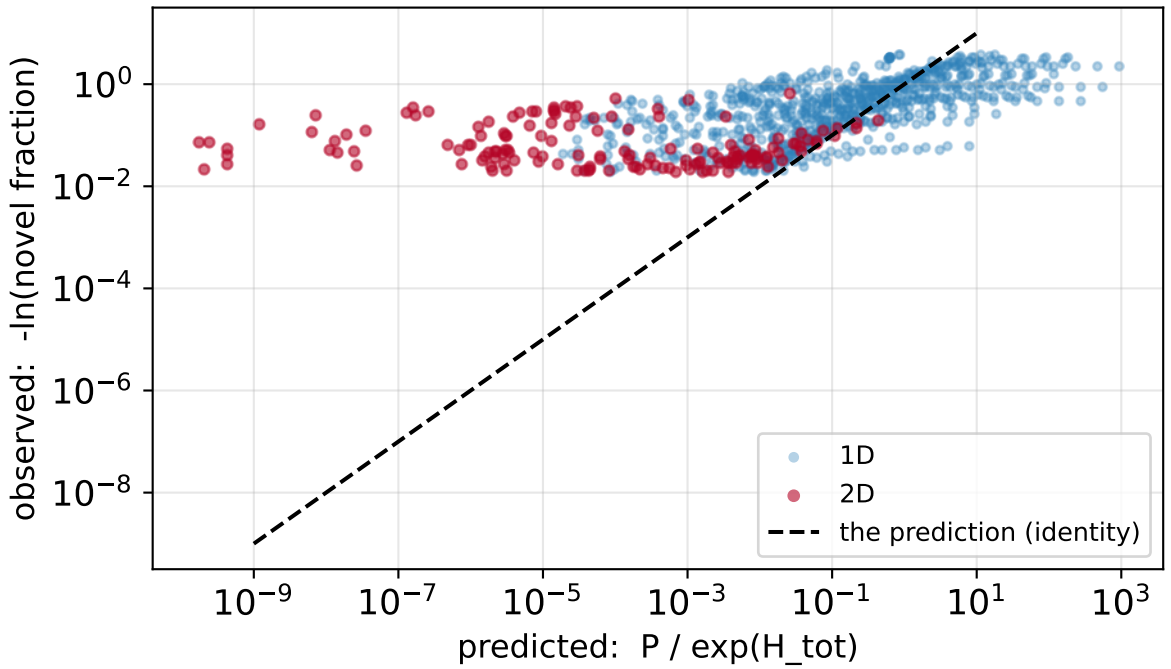
At n=16 that is k=11, at H_tot = 10.11 nats of a possible 16 ln 3 = 17.6. At n=8 it is k=5. Neither is U=1, and nothing in the entropy curve knows where U=1 is -- which is the point: the block structure sets where a window can first be grammatical at all, and the entropy sets where it should actually sit.

4. The mechanism holds at the level of single samples

within every run, copies are the low-entropy trajectories



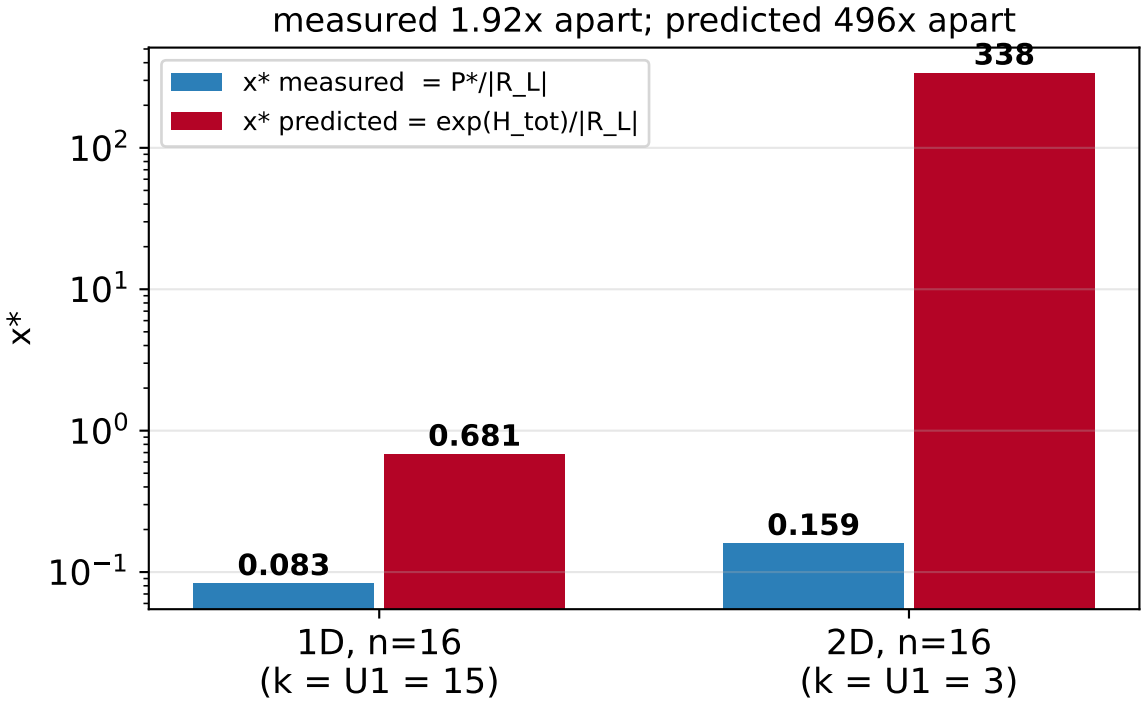
but the quantitative law misses by 2x in 1D, 600x in 2D

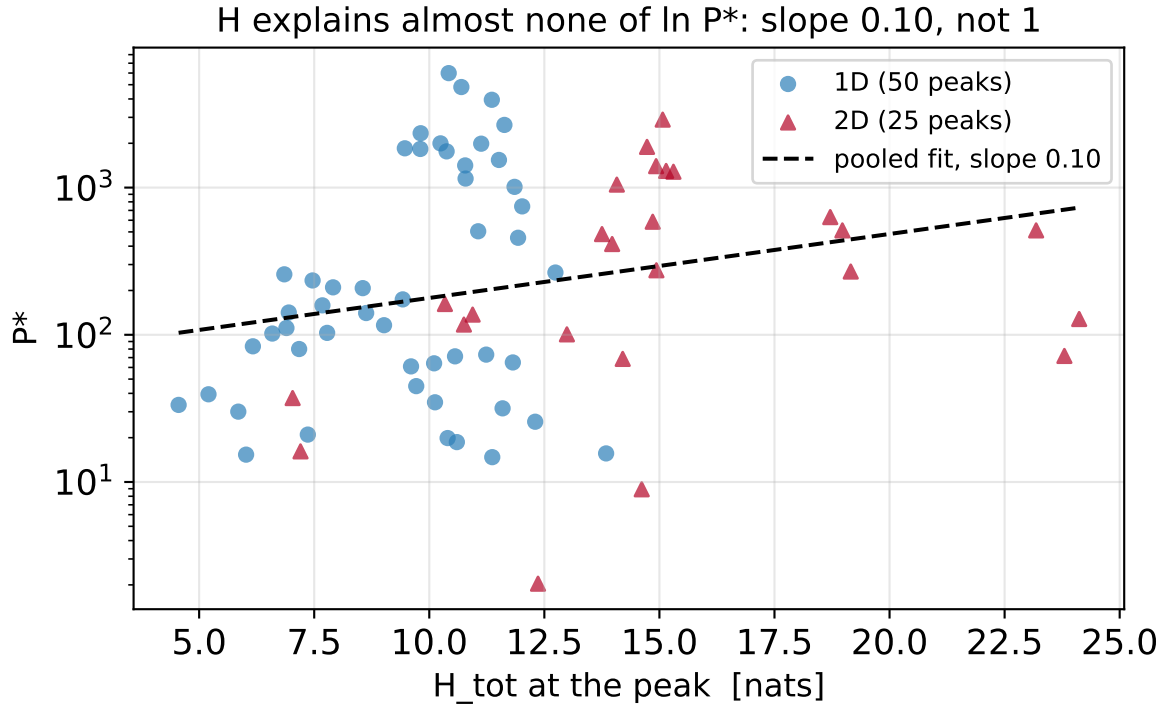


WHAT SURVIVES. Split each run's generations into the novel ones and the copies and compare their mean trajectory entropy. The copies come out lower in 84% of 1D runs, by a median 1.75 nats, and in 100% of 2D runs, by 7.42. The sampler reproduces a training string exactly when the windows pinned it, and pinning is visible as missing entropy. That is a mechanism, not a correlation: it is measured inside the run that produced the sample.

WHAT DOES NOT. Turning it into a count fails. If the machine spread its mass over exp(H_tot) strings, novelty would be exp(-P/exp(H_tot)) -- the dashed line. It is off by 2.06x in 1D with a 15x spread and by 597x in 2D with a 171x spread. exp(H_tot) is not the size of an effective support; the distribution is far too uneven for its entropy to count anything.

5. x* before and after: the budget makes the gap 258x worse





BEFORE, the same bracketed-peak eval both decks use, but for the first time with one instrument on both sides rather than two runs: 1D n=16 at U1=15 gives x* = 0.083 (6 cells), 2D n=16 gives 0.159 (all 25 bracketed L=2 cells). Gap 1.92x. The 2D figure reproduces the published 0.193 to 1.2x as a population median, but cell by cell the peak estimate scatters 4.8x -- 2048 generations and a 3-point parabola on a doubling ladder -- so it is a population number and no single cell should be quoted from it.
AFTER, substituting the entropy prediction x* = exp(H_tot)/|R_L|: 1D 0.68, 2D 337.6. Gap 496x -- 258 times worse than the thing it was meant to explain. And as a mere collapse variable H fails too: ln P* = 0.100 H + 4.18 pooled, residual 5.5x, and adding a dimension term still leaves a 2.1x 2D offset. The entropy budget joins coverage and table size on the refuted pile.

6. A novel generation and a copy, reveal by reveal

v=3, s=2, L=4 (n=16 sites), m=2; P=8192 training strings; window k=11; alphabet {0,1,2}, "." = still masked

NOVEL (H_tot = 11.27 nats)

```
................  site  7 -> 1   H = 0.89
.......1........  site 14 -> 2   H = 0.66
.......1......2.  site  9 -> 2   H = 0.91
.......1.2....2.  site  0 -> 2   H = 0.62
2......1.2....2.  site  6 -> 2   H = 0.72
2.....21.2....2.  site 10 -> 2   H = 0.89
2.....21.22...2.  site  2 -> 2   H = 0.67
2.2...21.22...2.  site 13 -> 1   H = 1.01
2.2...21.22..12.  site  1 -> 2   H = 1.06
222...21.22..12.  site  4 -> 1   H = 0.69
...  final 2221122112222121
```

grammatical: yes **novel: yes**

COPY (H_tot = 10.04 nats)

```
................  site 14 -> 1   H = 0.66
..............1.  site  8 -> 1   H = 0.89
........1.....1.  site  0 -> 1   H = 0.62
1.......1.....1.  site 12 -> 2   H = 0.56
1.......1...2.1.  site  7 -> 2   H = 0.71
1......21...2.1.  site  4 -> 2   H = 0.58
1...2..21...2.1.  site 10 -> 2   H = 0.92
1...2..21.2.2.1.  site 15 -> 2   H = 0.00
1...2..21.2.2.12  site 13 -> 1   H = 1.04
1...2..21.2.2112  site  9 -> 2   H = 0.00
...  final 1212221212222112
```

grammatical: yes **novel: NO**

Each row is one reveal: the string so far, the site the sampler chose uniformly among those still masked, the token it drew, and H, the entropy in nats of the posterior it drew from. H_tot is the sum over all 16 reveals -- the quantity every plot in this deck is built on. The first 10 reveals are shown.

The contrast is the mechanism in miniature, and it is worth being exact about its size: these two trajectories differ by 1.2 nats, against a median gap of 1.75 over the 781 1D runs. What makes the difference is reveals the windows have already pinned -- H = 0.00 exactly, where the k=11 windows admit one continuation and no other. The copy hits two in its first ten; the novel generation hits none, and every reveal with two or more live tokens is a place it can leave the training set without leaving the grammar.

7. Where this leaves the 1D/2D gap

ANSWERED. Why success peaks at k=11 rather than at the root-reaching U1=15. Widening the window from k=3 to k=17 removes 3.61 of 11.82 nats from the generation while grammaticality rises 0.06 -> 0.78 and novelty falls 1.00 -> 0.52. Copies are the low-entropy trajectories -- 1.75 nats below the novel ones in 1D, 7.42 in 2D, in 84% and 100% of runs. So the window is bought with exactly the quantity novelty needs, and the optimum is where the exchange stops paying. Nothing in the entropy curve marks U=1; the block structure says where a window can first be grammatical, the entropy says where it should sit.

STILL OPEN, AND NOW WITH A THIRD REFUTATION. $x^* = P^*/|R_L|$ measured head-to-head on one instrument: 1D 0.083, 2D 0.159, 1.92x apart. Substituting $\exp(H_{tot})/|R_L|$ gives 0.68 and 337.6 -- 496x apart, 258x worse. Pooled, $\ln P^* = 0.100\ H + 4.18$ with a 5.5x residual, so H carries almost none of $\ln P^*$. Coverage, table size and now entropy have all failed to collapse the dimensions.

WHAT THE FAILURE ITSELF SAYS. $\exp(H_{tot})/|R_L|$ is useless as a constant but reads cleanly as a regime: it is 0.68 in 1D and 338 in 2D. The 2D machine spreads its mass over vastly more strings than the grammar contains, so its peak is grammar-limited (peak success 0.022); the 1D machine is spread comparably to the grammar, so its peak is novelty-limited (peak success 0.49). Two different binding constraints is a reason no single ratio should collapse them, and it is measured rather than assumed. I am not calling it the explanation -- it is a reinterpretation of a number that failed its own test, and it predicts nothing yet.

NEXT. The AdaptiveWindow machine already in the repo picks k per site and per step off an entropy floor, which is what page 3 says the optimum actually is -- but it has only ever been scored on grammaticality, never on novel-and-grammatical. Scoring it is the measurement these three decks have been converging on.